

SAUDI GEO-CONSULT ENGINEERING CO.

Geotechnical Investigation Report

Report Title	Geotechnical Investigation Report - Soil & Foundation Recommendations
Project	Riyadh Metro-Adjacent Mixed Development
Report No.	SGC-RYD-GEO-2024-007
Date of Fieldwork	18-25 November 2023
Date of Report Issue	10 January 2024
No. of Boreholes	12 (depths ranging 15m - 30m)
Prepared For	Riyadh Metro-Adjacent Mixed Development — Main Contractor

Executive Summary

Subsurface conditions comprise 2-4m of fill/made ground overlying medium-dense to dense silty sand, transitioning to weathered limestone bedrock at approximately 9-14m depth. Groundwater was not encountered within the investigated depths. Allowable bearing capacity for shallow pad foundations is assessed at 250 kPa at 2.5m founding depth; piled foundations bearing on competent limestone are recommended for the tower structures exceeding 15 storeys.

Recommendations

1. Adopt bored cast-in-situ piles (600-900mm dia.) socketed 3m minimum into competent limestone for tower footprints.
2. Shallow foundations acceptable for podium/retail structures subject to ground improvement of upper fill layer.
3. Sulfate content in soil samples indicates Class DS-2 exposure — use sulfate-resisting cement (SRC) for all substructure concrete.